



Lab Task(s) Report – Lab 01

Name:

Muhammad Abu Bakar

NU ID:

24K-0826

January 28th, 2026

—

Syllabus:

Database Systems Lab

Syllabus Code: CL-2005

—

Instructor:

Sir Osama Tahir

Q1)**Mandatory:**

- Employee Id
- Employee Name
- Employee Department
- Contact Details

Optional:

- Employee Salary
- Position

Q2)

Assuming a scenario of a Bank Management System an incorrect attribute that might be treated as an entity would be Employee Address

Q3)**Entities:**

- Patient
- Nurse
- Wards
- Staff
- Doctors

Exclusions: Cafeteria Menu, Parking as they are irrelevant and focus should be on patient care.

Q4)

A Mini World of a Restaurant, the attributes of 'Chef Shift Hours', 'Fish Weight' should not be stored, as they can be calculated from other data, would cause redundancy and are better if they are calculated at the time they are required.

Q5)

Simple:

- Account Holder's Name
- Account Type
- Account Creation Date

Derived:

- Balance
- Account Age
- Total Withdrawals

Q6)

Attribute Name	Data Type
Book ISBN Number	Integer (this would be the Primary Key)
Book Name	Varchar
Book Price	Decimal
Genre	Varchar
Available To Issue	Boolean

Q7)

- Membership Active
- Is Paid
- Premium Membership Status
- Has Juice Bar Subscription
- Uses Trainer

Q8)

Entities:

- Courses
- Faculty

Attributes:

- Course Name
- Course Code/ID
- Credit Hour
- Has Pre-Requisite
- Number of Sections
- Number of Enrolled Students

- Faculty Name
- Faculty ID
- On-Board Status
- Faculty Contact Number

Q9)

Because the focus is on patient and treatment information and Parking Lot would be irrelevant.

Q10)

Entities:

- Aircraft
- Staff
- Airport
- Passenger

Attributes:

- Aircraft ID
- Model
- Maintenance Scheduled Date
- Manufacturer

- Staff ID
- Staff Name
- Staff Salary

- Airport ID
- Airport Name
- Location

- Passenger Name
- Passport Number
- Nationality
- Contact Info

Q11)

Attribute Name	Data Type	Reason/Justification
Supplier ID	Integer (Primary Key)	It must be unique, numerical sequence is used for unique identification, and an integer-based key was chosen over an alpha numeric based key so that querying and indexing is faster
Supplier Name	Varchar	As the Supplier's Name might vary in length, using Varchar would be beneficial as Varchar is dynamic so it would only use the storage space required for the actual text
Point of Contact (POC)	Varchar	POC's name must be stored as a string
POC Contact Number	Varchar	Varchar permits the usage of +, -, and parenthesis. A phone number should not be an integer, so that we can preserve leading zero's
Credit Limit	Decimal	It must be stored as a numeric value, as currency requires exact precision. Unlike float or double it avoids rounding errors, ensuring financial accuracy.

Q12)

For a Student/University Mini World if we store the 'Student's Name' is stored in each and every Course enrollment record this would cause data redundancy as every time a student enrolls in a course their name is repeated, instead we would be better off by linking the table/entity to a Student Entity which would contain the 'Student's Name' and then we should store the 'Student ID' as a foreign key in the Course Enrollment Table.

Professor's Email if it is repeated in every class instead of being stored in the Professor/Faculty table.